

Weighted mean

A two-value model

A weighted mean gives each value an assigned weight.

x_1 and x_2 are the two values; w_1 and w_2 are their weights.

Both weights are nonnegative, and $w_1 + w_2$ must be positive.

$$m = (w_1 * x_1 + w_2 * x_2) / (w_1 + w_2)$$

When the weights are equal, this is the ordinary mean.

Multiplying both weights by the same positive factor leaves m unchanged.

Worked example

Values 8 and 10, with weights 3 and 1

Value	Weight	Value x weight
8	3	24
10	1	10

$$m = (3 * 8 + 1 * 10) / (3 + 1) = 34 / 4 = 8.5$$

The value 8 has three times the weight of 10.

The result 8.5 is closer to 8. Equal weights would instead give 9.

